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# AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 304 — Past Questions & Materials

**by Renance**

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Course: CSC 304

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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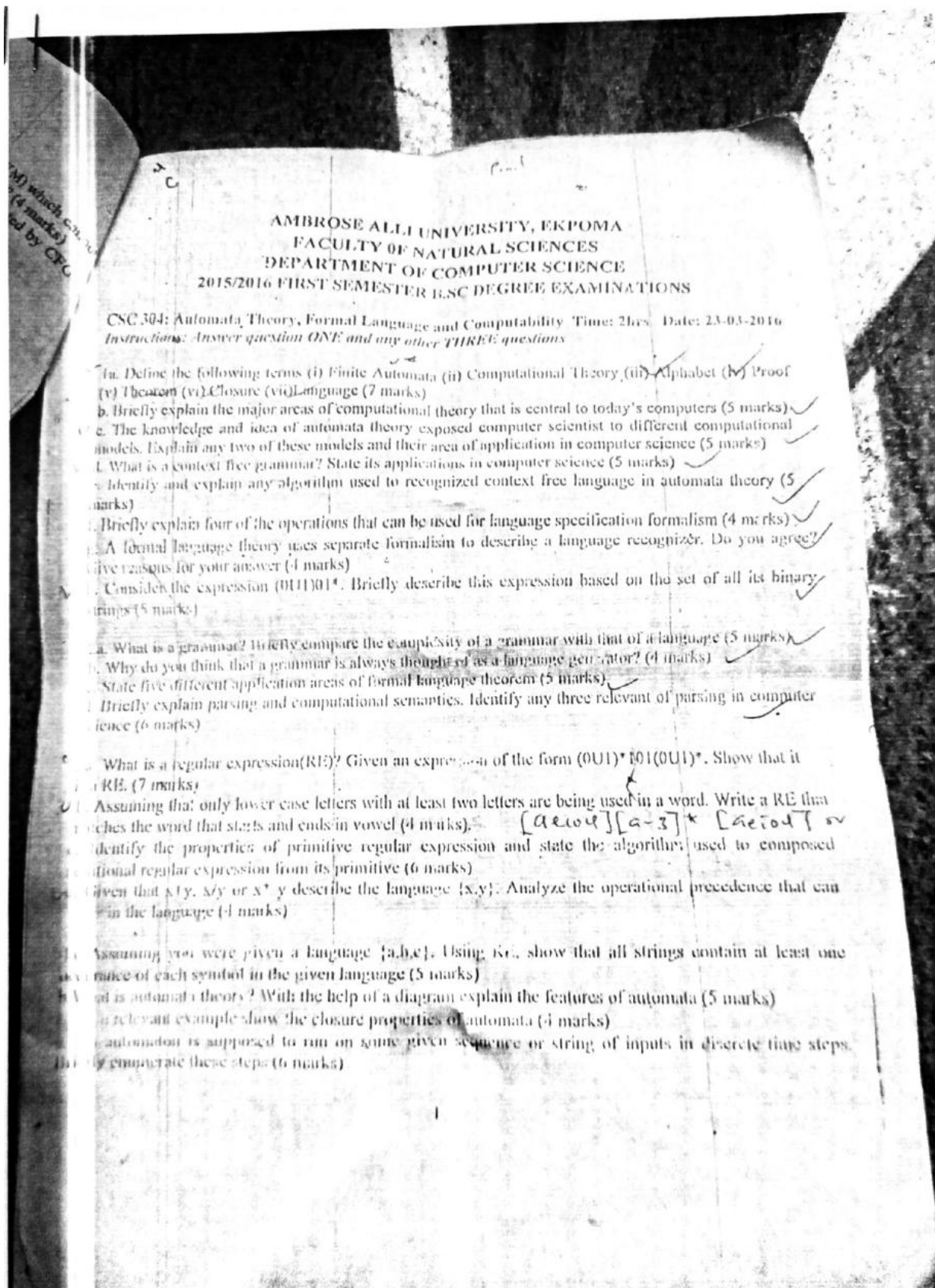
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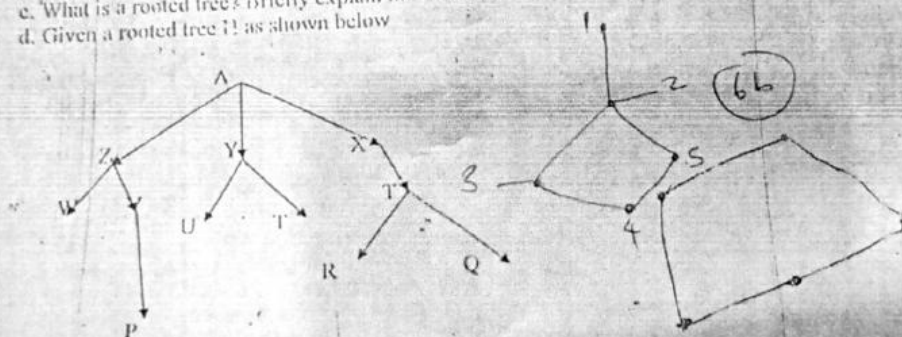
# Paper 1 of 1 — CSC 304

Level 300 • First Semester • PastQuestion

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- 5a. Briefly describe the Chomsky Hierarchy and explain the two types of this hierarchy (6 marks)  
 b. What are the two major extension and variations on Chomsky's original hierarchy of formal gr (4 marks)  
 ✓ c. List the alphabets of either a JAVA or C++ programming language (4 Marks)  
 ✓ d. With relevant examples, describe any three set operations that can combine two sets to form a third (4 marks)
- 6a. Define (i) Graph (ii) Planar Graph (iii) Subgraph (iv) Node (v) Connected graph (vi) Digraph (6 marks)  
 b. Draw a complete graph with five vertices (4 marks)  
 c. What is a rooted tree? Briefly explain the roles of tree in computer science (4 marks)  
 d. Given a rooted tree as shown below

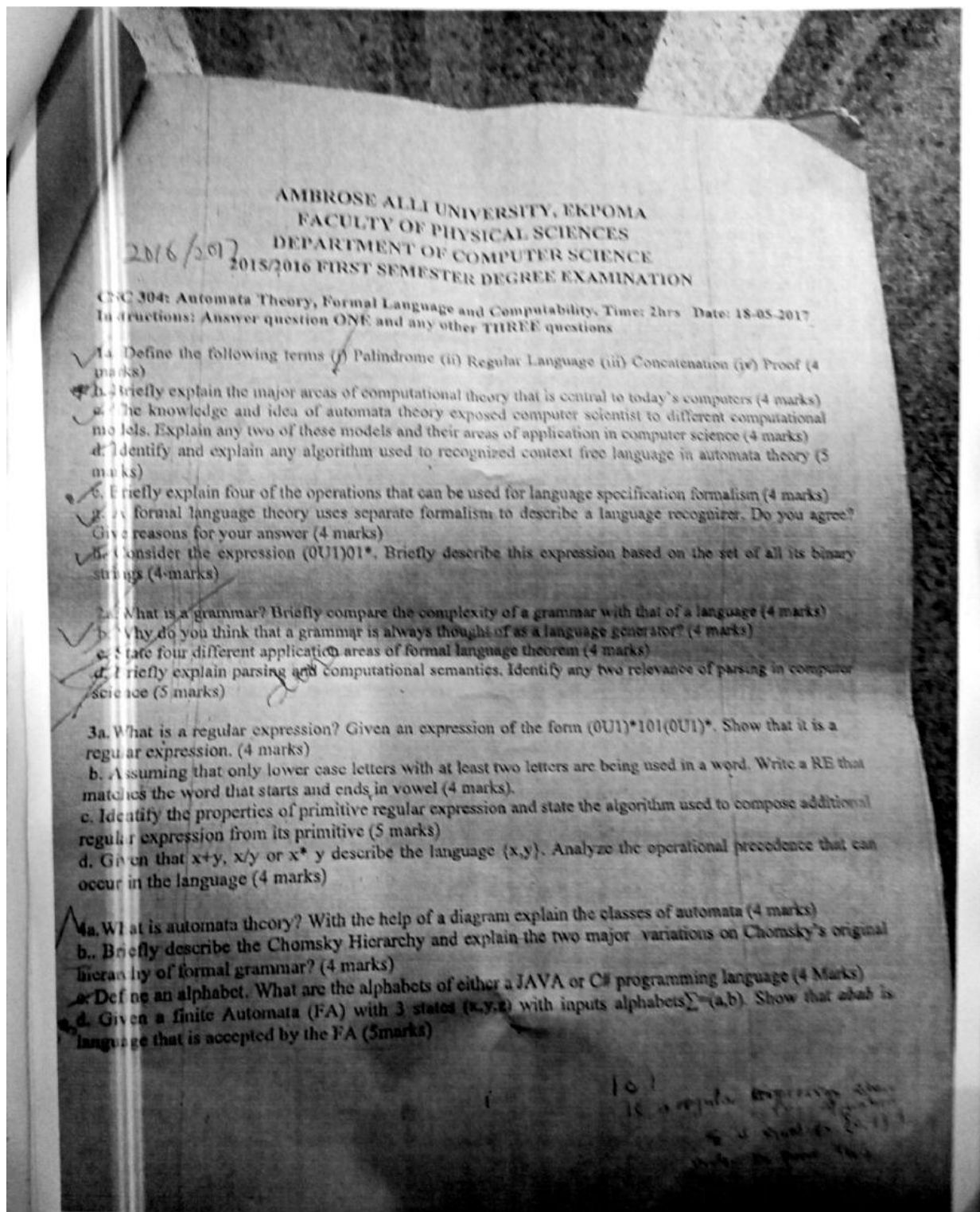


- What is the root of T? (1 mark)
- Identify the parent of the node w (1 mark)
- Identify the children of the node x (1 mark)
- Identify the children of node w (1 mark)
- What is the level of each node? (1 mark)
- Which nodes are in the sub-tree rooted at z (1 mark)

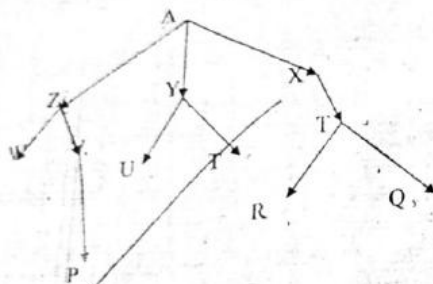
vi) The sub tree rooted at z contains the nodes, z, w, v, p, u, t, r, q.

$\gamma, \alpha, \beta, \dots$

$w, w_2, \dots, w_k: k \geq 0$  and  $w_k \neq w_{k+1}$



- 5a. Similar to a FA but with an unlimited and unrestricted memory is a Turing Machine(TM) which can solve certain problems that a real computer can do. What is a TM and how can it be defined? (4 marks)
- b. Give a formal definition of a Context Free Grammar (CFG). If  $L_1$  be the language generated by  $C_1$  and  $L_2$  be the language generated by regular expression. Show that  $L_1 = L_2$  (4 marks)
- c. What is a rooted tree? Briefly explain the roles of tree in computer science (3 marks)
- d. Given a rooted tree T as shown below



- What is the root of T? (1 mark)
- Identify the parent of the node w (1 mark)
- Identify the children of the node x (1 mark)
- Identify the children of node w (1 mark)
- What is the level of each node? (1 mark)
- Which nodes are in the sub-tree rooted at z (1 mark)

- 6 a). When is a language said to be regular? If  $L_1$  and  $L_2$  are regular languages, show that  $L_1 + L_2$  and  $L_1 L_2$  are regular languages. (5 marks)
- b) What is an irregular language? Is the language  $\{0^n 1^n : n > 0\}$  regular or irregular? Justify your claim. (4 marks)
- c) State the Pumping Lemma. The Pumping lemma relates the size of strings accepted with the number of states in a DFA. Briefly states what the Pumping lemma says about the above assertion. (5 marks)
- d) When do we say that a problem is effectively solvable? Consider the problem of finding roots of a quadratic equations of the form  $Y = ax^2 + bx + c$ , justify that it is effectively solvable. (3 marks).

**AMBROSE ALLI UNIVERSITY, EKPOMA**  
**FACULTY OF PHYSICAL SCIENCES**  
**DEPARTMENT OF COMPUTER SCIENCE**  
**2019/2020 FIRST SEMESTER DEGREE EXAMINATIONS**

**CSC 304: Automata Theory, Formal Language and Computability**

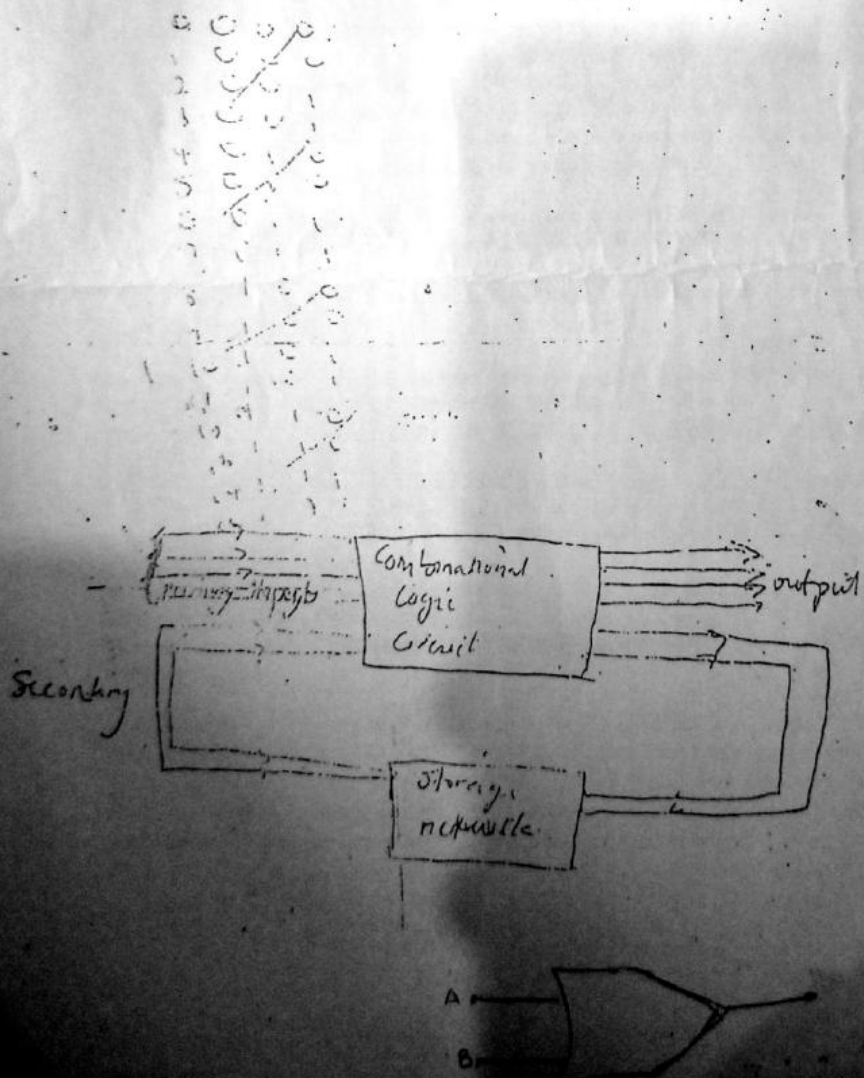
**Date: 06/03/2021**

**Instructions: Answer Question one and any other three Questions.**

**Time: 2 Hours**

- 1a. Define the following terms (i) Symbol (ii) Alphabet (iii) Language (iv) Theorem (v) String (5 marks)
- b. What is a string? Determine the set of all strings over the alphabet  $\Sigma = \{0,1\}$  that start with 0 and design its state machine (5 marks)
- c. Using the state machine in 1b, practically show if the following strings are accepted or not and explain why. (i) 0111 (ii) 011 (iii) 10011101 (iv) 00111010110 (v) 010001 (5 marks)
- d. What are finite automata? Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start and end with "a"}, w \in \Sigma^*\}$ ,  $\Sigma = \{a, b\}$ . You are required to design its transition diagram. (3 marks)
- e. Given that  $L = \{w/w \text{ start and end with "x"}, w \in \{x, y\}^*\}$ . Design the finite automata that show the desired state and undesired state of the machine (5 marks)
- f. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start and end with "n"}, w \in \Sigma^*\}$ ,  $\Sigma = \{n, m\}$ . Design the DFA and the NFA. (6 marks)
- 2a. What is formal Grammar? How is it different from formal language? (5 marks)
- b. Design a formal grammar and language of your choice and check for any syntactic error when parsing the language. (7 marks)
- c. Using relevant example describe any five different operations on formal languages (5 marks)
- 3a. What is Regular Expression (RE)? With relevant example explain the properties of RE (6 marks)
- b. Consider the regular expression  $(0u1)01^*$ . Briefly describe this expression based on the set of all its binary strings (5 marks)
- c. Identify the properties of primitive regular expression and state the algorithm used to compose additional regular expression from its primitive (6 marks)
- 4a. Briefly describe the following: (i) Complexity theory (ii) Computability theory (iii) Automata theory (6 marks)
- b. Design an ambiguous grammar and briefly show its ambiguity using any language of your choice. (6 marks)
- c. Design an algorithm that can be used to describe a language from a grammar. (5 marks)
- 5a. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start with "4" and end with "5"}, w \in \Sigma^*\}$ ,  $\Sigma = \{4, 5\}$ . Design its DFA. (5 marks)
- b. What is a context free grammar? With relevant example differentiate between terminals and non terminals of a grammar. (5 marks)
- c. Design a grammar that can be used to parse the language  $xyyaab$  and show its parsing. (7 marks)
- 6a. What is a syntax tree? Given that  $M \rightarrow M+M \mid M*M \mid \text{id}$ , generate the syntax tree for the strings  $x+y*z$ . (Note:  $M = \text{Expression}$  and  $\text{id} = \text{identifier}$ ). (6 marks)
- b. Practically show if ambiguity exist in the production in question 6a or not. (6 marks)
- c. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start with "y" and end with "z"}, w \in \Sigma^*\}$ ,  $\Sigma = \{y, z\}$ . Design its NFA. (5 marks)

- 5(a) Define (i) Fan-in (ii) Fan-out (iii) MOS technologies (iv) Bipolar technologies  
(v) Outline and explain the technologies used in Bipolar technologies
- (b) What are integrated circuits (ICs)? Outline the various classes of IC and the possible applications of each class.
- (c) What is a Decoder? State its uses. Draw the truth table and the diagram for a 3-to-8 line decoder.
- 6(a) Briefly explain the operation of registers, and identify the various ways to transfer information in and out of a register.
- (b) Define a microprocessor. State its characteristics and explain how it executes each instruction given to it.
- (c) A micro-computer comprises of 3 major sub-systems. Identify and explain briefly each of these sub-systems.



AMBROSE ALLI UNIVERSITY, EKPOMA  
FACULTY OF PHYSICAL SCIENCES  
DEPARTMENT OF COMPUTER SCIENCE  
2017/2018 FIRST SEMESTER DEGREE EXAMINATION

**CSC 304: Automata Theory, Formal Language and Computability** Time: 2hrs Date: 19-04-2018  
Instructions: Answer question ONE and any other THREE questions

1. (a). Define the following concepts (i) Concatenation (ii) Regular Language (iii) Formal language (iv) Planar Graph (v) Theorem (vi) Syntax Tree (vii) Palindrome (7 marks)  
(b). Automata theory deal with definitions and properties of different types of computational models. Explain three examples of these models. (3 marks)  
(c). Is it true that actual computer is different from finite automata? Justify your claim with superior argument (3 marks)  
(d). State Kleene's theory. Why is it important in automata theory? (4 marks)  
(e). (i) What do you understand by automata theory? (ii) State any three classes of automata. (5 marks)  
(f). The theory of computation can be sub-divided into different areas. Explain any three of these areas and state why the sub-division is necessary in automata theory (4 marks)
2. (a). Define language. List and explain with examples the operations on language. (6 marks)  
(b). Given a regular language X and Y over the same alphabet, show that XUY is also a regular language. (6 marks)  
(c). State the pumping lemma theorem. Explain briefly how you can use a pumping lemma to prove that a language is not regular. (6 marks)
3. (a). What is grammar? How is it different from a language? Outline briefly how you can use a grammar to describe a language by generating each string of that language (7 marks)  
(b). Given that  $L_1$  is a context-free language (CFL), and  $L_2$  is the language generated by the regular expression  $(a+b)^*$ . Show that  $L_1 = L_2$ . (5 marks)  
(c). What is context free grammar (CFG)? Identify 4 areas where it is applied in computer science. (6 marks)
4. (a). What is Pushdown Automata? How is it different from Non-Deterministic Finite Automata? State some of its equivalence with CFG (7 marks)  
(b). Given that  $L_1$  and  $L_2$  are regular language, show that the concatenation and star operations of this language are also regular (5 marks)  
(c). When do we say a problem is effectively solvable? Given that  $x^2 + y^2 + z^2 = 0$ , justify that it is effectively solvable (6 marks).
5. (a). Similar to a FA but with an unlimited and unrestricted memory is a Turing machine (TM) which cannot solve certain problems that a real computer can do. What is TM and how can it be defined (6 marks)  
(b). Outline any 4 differences between a FA and a TM and identify any 4 tricks of a TM (7 marks)  
(c). Given a Finite Automata (FA) with 3 states (x, y and z) with inputs alphabets  $\Sigma = \{a, b\}$ . Show that abab is a language that is accepted by the FA (5 marks)

See if words are final string

See if words are final string

is a simple idealized machine which recognizes pattern within strings using some set of characters (or symbols)

2081731754 Mitchell Adusi 2/500

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**FACULTY OF PHYSICAL SCIENCES**  
**DEPARTMENT OF COMPUTER SCIENCE**  
**2018/2019 FIRST SEMESTER DEGREE EXAMINATIONS**

**CSC 304: Automata Theory, Formal Language and Computability** Date: 25/5/2019  
**Instructions: Answer Question one and any other three Questions.** Time: 2 Hours

1a. Define the following terms (i) Symbol (ii) Alphabet (iii) Language (iv) Theorem (v) Closure (5 marks)

b. What is a string? Determine the set of all strings over the alphabet  $\Sigma = \{0,1\}$  that start with 1 and design its state machine (5 marks)

c. Using the state machine in 1b, practically show if the following strings are accepted or not and explain why. (i) 111 (ii) 011 (iii) 10011101 (iv) 00111010110 (6 marks)

\* d. What are finite automata? Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start and end with "a"}\}$ ,  $W \in \Sigma^*$ ,  $\Sigma = \{a, b\}$ . You are required to design its transition diagram. (4 marks)

e. What is a formal language? Identify six different operations that are frequently used in the study of formal languages (5 marks)

f. Given that  $L = \{w/w \text{ start and end with "a"}\}$ ,  $W \in \{a, b\}^*$ . Design the finite automata that show the desired state and undesired state of the machine (5 marks)

2a. Assuming you were given a language  $\{a, b, c\}$ . Using RE, show that all strings contain at least one occurrence of each symbol in the given language (5 marks)

b. What is automata theory? With the help of a diagram explain the features of automata (5 marks)

c. With relevant example show the closure properties of automata (4 marks)

d. An automaton is supposed to run on some given sequence or string of inputs in discrete time steps. Briefly enumerate these steps (6 marks)

3a. Briefly describe the Chomsky Hierarchy and explain the two types of this hierarchy (6 marks)

b. What are the two major extension and variations on Chomsky's original hierarchy of formal grammar? (4 marks)

c. Using computational theory in automata systems, prove that if  $yx^2=0$ , then  $x=0$  (4 marks)

d. With relevant example, describe any three set operations that can combine two sets to form a third one (6 marks)

4a. What is a grammar? Briefly compare the complexity of a grammar with that of a language (5 marks)

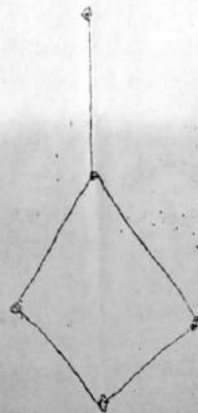
b. Why do you think that a grammar is always thought of as a language generator? (4 marks)

c. State five different application areas of formal language theorem (5 marks)

d. Briefly explain parsing and computational semantics. Identify any three relevant of parsing in computer science (6 marks)

5a. What is Regular Expression (RE)? With relevant example, explain the properties of RE (6 marks)

- b. Assuming that only lower case letters with at least two letters are being used in a word. Write RE that matches the word that starts and ends in vowel (4 marks).
- c. Identify the properties of primitive regular expression and state the algorithm used to composed additional regular expression from its primitive (6 marks)
- d. Given that  $x+y$ ,  $x/y$  or  $x^*y$  describe the language  $\{x,y\}$ . Analyze the operational precedence that can occur in the language (4 marks)
- 6a. Define (i) Graph (ii) Planar Graph (iii) Subgraph (iv) Node (v) Connected graph (vi) Digraph (6 marks)
- b. Draw a complete graph with five vertices (4 marks)
- c. What is a rooted tree? Briefly explain the roles of tree in computer science (4 marks)
- d. Briefly explain the three major areas of computational theory that is central to today's computers (6 marks)



vertices =  $\{1, 2, 3, 4, 5\}$   
 edges =  $\{(1,2), (2,3), (4,5), (5,2), (5,4)\}$

**AMBROSE ALLI UNIVERSITY, EKPOMA**  
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**2020/2021 FIRST SEMESTER DEGREE EXAMINATIONS**

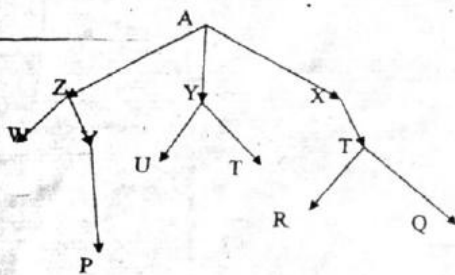
CSC 304: Automata Theory, Formal Language and Computability  
 Instructions: Answer Question one and any other three Questions.

Date: 20/12/2021

Time: 2 Hours

- 1a. Define the following terms (i) Symbol (ii) Alphabet (iii) Language (iv) Theorem (v) String (5 marks)
- b. Determine the set of all strings over the alphabet  $\Sigma = \{0,1\}$  that start with 0 and design its state machine (4 marks)
- c. Using the state machine in 1b, practically show if the following strings are accepted or not. (i) 1100111 (ii) 011 (iii) 0010011101 (iv) 111010111 (4 marks)
- d. What are Finite Automata (FA)? Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start and end with "a"}, w \in \Sigma^*\}$ ,  $\Sigma = \{a,b\}$ . You are required to design its transition diagram. (3 marks)
- e. What is a Turing Machine (TM)? Construct a TM that accept the language of aba over  $\Sigma = \{a,b\}$  and show the transition diagram (5 marks)
- f. Design a TM for the language  $L = \{0^n 1^n 2^n\}$  where  $n > 1$  (5 marks)
- 2a. What is a syntax tree? Given that  $\Lambda \rightarrow \Lambda - \Lambda \mid \Lambda * \Lambda \mid \Lambda^* \Lambda \mid \text{id}$ , generate the syntax tree for the strings  $x y z$  and  $a^* b^* c$ . (Note:  $\Lambda$  = Expression and id = identifier). (8 marks)
- b. Practically show if ambiguity exist in the productions in question 2a or not. (6 marks)
- c. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start with "y" and end with "z"}, w \in \Sigma^*\}$ ,  $\Sigma = \{y, z\}$ . Design its NFA. (4 marks)
- 3a. What is formal Grammar? How is it different from formal language? (6 marks)
- b. Design a formal grammar and language of your choice and check for any syntactic error when parsing the language. (7 marks)
- c. Using relevant example describe any five different operations on formal languages (5 marks)
- 4a. What is Regular Expression (RE)? With relevant example explain the properties of RE (7 marks)
- b. Design the transition diagram for the TM in question 1f (5 marks)
- c. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start and end with "x"}, w \in \Sigma^*\}$ ,  $\Sigma = \{x, y\}$ . Design the DFA and the NFA. (6 marks)
- 5a. Briefly describe the following: (i) Complexity theory (ii) Computability theory (iii) Automata theory (6 marks)
- b. Design an ambiguous grammar and briefly show its ambiguity using any string of your choice. (6 marks)
- c. As a computing system expert, design a FA for a computing machine at Lekki toll gate that can only be open for a driver that successfully paid #70 as toll gate fee (6 marks).
- 6a. Given that  $m = (Q, \Sigma, \delta, q_0, F)$ ,  $L = \{w/w \text{ start with "4" and end with "5"}, w \in \Sigma^*\}$ ,  $\Sigma = \{4, 5\}$ . Design its DFA. (5 marks)
- b. What is a context free grammar? With relevant example differentiate between terminals and non terminals of a grammar. (6 marks)
- c. Design a grammar that can be used to parse the string  $acxyaab$  and show its parsing. (7 marks)

- 5a. Similar to a FA but with an unlimited and unrestricted memory is a Turing Machine(TM) which cannot solve certain problems that a real computer can do. What is a TM and how can it be defined? (4 marks)
- b. Give a formal definition of a Context Free Grammar (CFG). If  $L_1$  be the language generated by CFG, and  $L_2$  be the language generated by regular expression. Show that  $L_1 = L_2$  (4 marks)
- c. What is a rooted tree? Briefly explain the roles of tree in computer science (3 marks)
- d. Given a rooted tree H as shown below *→ is a tree with a root*



- What is the root of H? (1 mark)
- Identify the parent of the node w (1 mark)
- Identify the children of the node x (1 mark)
- Identify the children of node w (1 mark)
- What is the level of each node? (1 mark)
- Which nodes core in the sub-tree rooted at z (1 mark)

(a). When is a language said to be regular? If  $L_1$  and  $L_2$  are regular languages, show that  $L_1 + L_2$  and  $L_1 L_2$  are regular languages. (5 marks)

(b) What is an irregular language? Is the language  $\{0^n 1^n : n > 0\}$  regular or irregular? Justify your claim. (4 marks)

(c). State the Pumping Lemma. The Pumping lemma relates the size of string accepted with the number of states in a DFA. Briefly states what the Pumping lemma says about the above assertion. (5 marks)

(d) When do we say that a problem is effectively solvable? Consider the problem of finding roots of a quadratic equations of the form  $Y = ax^2 + bx + c$ , justify that it is effectively solvable. (3 marks).

*Alphabet is a finite set of symbols*